

NKPC HSA Bayesian State-Space Estimation

June 23, 2026

1 Data and Transformations

Raw data are read from `data/raw/`. Processed model-ready data are written to `data/processed/`. The HSA competition aggregator convention is

$$N_t^{model} = \frac{100 \log(N_t^{level}) - \overline{100 \log(N^{level})}}{10}.$$

Thus N_t^{model} , $\hat{N}_t$, and $\bar{N}_t$ are centered and measured in ten-log-point units. If a run metadata file records another `n_transform`, interpret the corresponding coefficients using that metadata.

2 Model Specifications

The CES benchmark is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + \kappa x_t + e_t.$$

The dynamic HSA specification is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + \kappa x_t - \theta \hat{N}_t + e_t.$$

The steady HSA specification is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + (\kappa_0 + \delta \bar{N}_t) x_t + e_t.$$

The both-effects HSA specification is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + (\kappa_0 + \delta \bar{N}_t) x_t - (\theta_0 + \gamma \bar{N}_t) \hat{N}_t + e_t.$$

The maintained HSA decomposition is

$$N_t = \hat{N}_t + \bar{N}_t, \quad \hat{N}_t = \rho_1 \hat{N}_{t-1} + \rho_2 \hat{N}_{t-2} + u_t, \quad \bar{N}_t = n + \bar{N}_{t-1} + \epsilon_t.$$

3 Unit and Covariance Conventions

Priors for κ , κ_0 , and δ are specified in physical units. Legacy Gibbs samplers internally use `KAPPA_SCALE=100` when the design matrix contains $x_t/100$. All posterior draws, tables, figures, SDDR calculations, and model-comparison summaries are reported in physical units. The default dynamic-HSA shock covariance allows only (e_t, ζ_t) correlation; diagonal and historical full covariance modes are available only through explicit run metadata. Because the default N transform is centered and divided by 10, the reported δ , θ , θ_0 , and γ coefficients are already effects for a ten-log-point movement in the corresponding N component. κ_0 and θ_0 are intercepts at average $\bar{N}_t$.

4 Coefficient Summary

Table 1 reports posterior means and 95% credible intervals for key parameters across all four models (baseline prior, full sample). Cells show *mean* [*ci_2.5*, *ci_97.5*].

Table 1: Posterior means and 95% credible intervals by model

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.640 [0.536, 0.747]	0.640 [0.537, 0.742]	0.617 [0.509, 0.719]	0.587 [0.480, 0.699]
kappa	-0.008 [-0.168, 0.150]	NaN	-0.004 [-0.163, 0.156]	NaN
kappa_0	NaN	-0.006 [-0.173, 0.157]	NaN	-0.003 [-0.165, 0.164]
delta	NaN	0.016 [-0.020, 0.050]	NaN	0.009 [-0.027, 0.044]
theta	NaN	NaN	-0.017 [-0.380, 0.562]	NaN
theta_0	NaN	NaN	NaN	0.244 [0.068, 0.477]
gamma	NaN	NaN	NaN	-0.006 [-0.045, 0.034]
rho_1	NaN	0.744 [0.370, 1.051]	0.899 [0.612, 1.112]	0.986 [0.767, 1.203]
rho_2	NaN	-0.026 [-0.309, 0.239]	0.015 [-0.179, 0.245]	-0.041 [-0.252, 0.168]
phi_1	0.632 [0.498, 0.768]	0.632 [0.499, 0.767]	0.632 [0.505, 0.753]	0.632 [0.494, 0.765]
lambda_ez	0.010 [-0.202, 0.218]	0.017 [-0.193, 0.224]	NaN	0.018 [-0.181, 0.215]
n	NaN	-0.022 [-0.084, 0.038]	-0.033 [-0.080, 0.021]	-0.001 [-0.053, 0.051]

5 Result Blocks

Results are organized by condition. Each section fixes a data specification, prior specification, and sample period, then reports the coefficient table, prior/posterior overlay by model, Savage-Dickey density ratios, time-varying κ_t path, and model-comparison output for that condition.

6 inv_markup: prior=baseline, period=full

Condition. data=inv_markup, prior=baseline, period=full, constraint=unrestricted (unrestricted).

6.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.6395	0.5380	0.7421
ces	kappa	0.0067	-0.2171	0.2283
hsa_steady	alpha	0.6390	0.5362	0.7415
hsa_steady	kappa_0	0.0101	-0.2171	0.2408
hsa_steady	delta	-0.0031	-0.0397	0.0335
hsa_dynamic	alpha	0.6150	0.5038	0.7199
hsa_dynamic	kappa	0.0115	-0.2174	0.2397
hsa_dynamic	theta	-0.0225	-0.3589	0.5775
hsa_full	alpha	0.6024	0.4854	0.7183
hsa_full	kappa_0	0.0357	-0.2097	0.2828
hsa_full	delta	-0.0020	-0.0401	0.0359
hsa_full	theta_0	0.0564	-0.3799	0.4720
hsa_full	gamma	-0.0061	-0.0458	0.0323

6.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

6.3 Savage-Dickey Density Ratio

note

No output available for this block.

6.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

6.5 Model Comparison

note

No output available for this block.

7 inv_markup: prior=baseline, period=pre_2008

Condition. data=inv_markup, prior=baseline, period=pre_2008, constraint=unrestricted (unrestricted).

7.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.6919	0.5807	0.7979
ces	kappa	-0.0198	-0.2164	0.1785
hsa_steady	alpha	0.6927	0.5801	0.7994
hsa_steady	kappa_0	-0.0181	-0.2179	0.1863
hsa_steady	delta	-0.0023	-0.0408	0.0351
hsa_dynamic	alpha	0.6130	0.4839	0.7455
hsa_dynamic	kappa	0.0362	-0.1768	0.2500
hsa_dynamic	theta	0.3459	-0.3403	0.7562
hsa_full	alpha	0.5897	0.4586	0.7201
hsa_full	kappa_0	0.0704	-0.1430	0.2815
hsa_full	delta	-0.0040	-0.0418	0.0339
hsa_full	theta_0	0.3563	0.1485	0.6061
hsa_full	gamma	-0.0033	-0.0418	0.0335

7.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

7.3 Savage-Dickey Density Ratio

note

No output available for this block.

7.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

7.5 Model Comparison

note

No output available for this block.

8 inv_markup: prior=baseline, period=post_2008

Condition. data=inv_markup, prior=baseline, period=post_2008, constraint=unrestricted (unrestricted).

8.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.5680	0.3216	0.8072
ces	kappa	0.1216	-0.2399	0.4806
hsa_steady	alpha	0.5755	0.3316	0.8191
hsa_steady	kappa_0	0.1177	-0.2371	0.4673
hsa_steady	delta	-0.0000	-0.0405	0.0406
hsa_dynamic	alpha	0.5747	0.3473	0.7931
hsa_dynamic	kappa	0.1916	-0.1801	0.5619
hsa_dynamic	theta	0.1644	-0.3164	0.6374
hsa_full	alpha	0.5708	0.3301	0.8030
hsa_full	kappa_0	0.1242	-0.2155	0.4812
hsa_full	delta	0.0005	-0.0381	0.0413
hsa_full	theta_0	0.1002	-0.3162	0.5035
hsa_full	gamma	-0.0002	-0.0386	0.0387

8.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

8.3 Savage-Dickey Density Ratio

note

No output available for this block.

8.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

8.5 Model Comparison

note

No output available for this block.

9 inv_markup: prior=baseline, period=exclude_covid

Condition. data=inv_markup, prior=baseline, period=exclude_covid, constraint=unrestricted (unrestricted).

9.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.6377	0.5310	0.7408
ces	kappa	0.0105	-0.2120	0.2343
hsa_steady	alpha	0.6381	0.5340	0.7396
hsa_steady	kappa_0	0.0081	-0.2205	0.2357
hsa_steady	delta	-0.0024	-0.0389	0.0343
hsa_dynamic	alpha	0.6173	0.5056	0.7208
hsa_dynamic	kappa	0.0093	-0.2204	0.2473
hsa_dynamic	theta	-0.0224	-0.3681	0.5725
hsa_full	alpha	0.5759	0.4635	0.6893
hsa_full	kappa_0	0.0753	-0.1580	0.3187
hsa_full	delta	-0.0036	-0.0418	0.0345
hsa_full	theta_0	0.3158	0.1032	0.5877
hsa_full	gamma	-0.0037	-0.0400	0.0337

9.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

9.3 Savage-Dickey Density Ratio

note

No output available for this block.

9.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

9.5 Model Comparison

note

No output available for this block.

10 inv_markup: prior=baseline, period=start_1988

Condition. data=inv_markup, prior=baseline, period=start_1988, constraint=unrestricted (unrestricted).

10.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.6219	0.4971	0.7416
ces	kappa	0.0124	-0.2316	0.2676
hsa_steady	alpha	0.6211	0.5013	0.7391
hsa_steady	kappa_0	0.0121	-0.2343	0.2724
hsa_steady	delta	-0.0006	-0.0391	0.0360
hsa_dynamic	alpha	0.6121	0.4941	0.7241
hsa_dynamic	kappa	0.0197	-0.2248	0.2730
hsa_dynamic	theta	0.1449	-0.2471	0.5250
hsa_full	alpha	0.5804	0.4470	0.7151
hsa_full	kappa_0	0.0585	-0.1989	0.3188
hsa_full	delta	-0.0006	-0.0380	0.0367
hsa_full	theta_0	0.1566	-0.1151	0.4233
hsa_full	gamma	-0.0008	-0.0404	0.0374

10.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

10.3 Savage-Dickey Density Ratio

note

No output available for this block.

10.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

10.5 Model Comparison

note

No output available for this block.

11 inv_markup: prior=baseline, period=end_2019

Condition. data=inv_markup, prior=baseline, period=end_2019, constraint=unrestricted (unrestricted).

11.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.6395	0.5369	0.7398
ces	kappa	0.0095	-0.2166	0.2311
hsa_steady	alpha	0.6390	0.5365	0.7444
hsa_steady	kappa_0	0.0111	-0.2157	0.2383
hsa_steady	delta	-0.0026	-0.0399	0.0351
hsa_dynamic	alpha	0.6167	0.5054	0.7194
hsa_dynamic	kappa	0.0080	-0.2237	0.2372
hsa_dynamic	theta	-0.0389	-0.3544	0.5649
hsa_full	alpha	0.6277	0.5239	0.7305
hsa_full	kappa_0	0.0072	-0.2190	0.2264
hsa_full	delta	-0.0030	-0.0395	0.0327
hsa_full	theta_0	-0.1649	-0.3908	0.0429
hsa_full	gamma	-0.0040	-0.0414	0.0349

11.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

11.3 Savage-Dickey Density Ratio

note

No output available for this block.

11.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

11.5 Model Comparison

note

No output available for this block.

12 labor_share_gap_hp: prior=baseline, period=full

Condition. data=labor_share_gap_hp, prior=baseline, period=full, constraint=unrestricted (unrestricted).

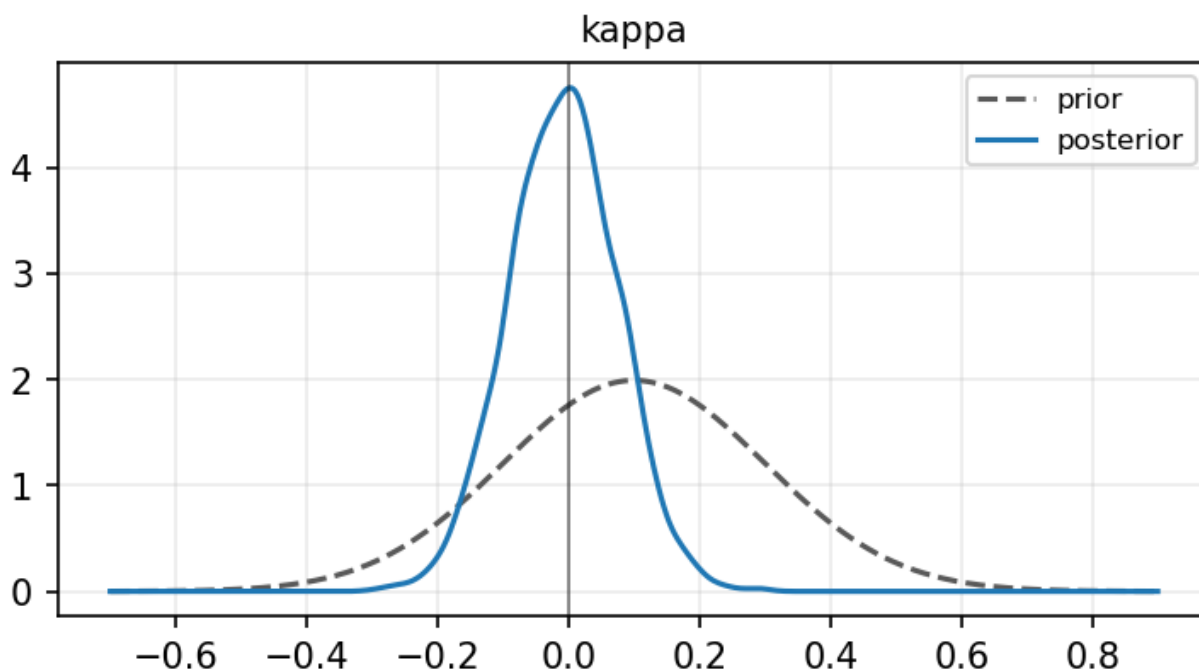
12.1 Coefficient Table

	ces	hsa_steady	hsa_dynamic	hsa_full
parameter				
alpha	0.640 [0.536, 0.747]	0.640 [0.537, 0.742]	0.617 [0.509, 0.719]	0.587 [0.480, 0.699]
kappa	-0.008 [-0.168, 0.150]	NaN	-0.004 [-0.163, 0.156]	NaN
kappa_0	NaN	-0.006 [-0.173, 0.157]	NaN	-0.003 [-0.165, 0.164]
delta	NaN	0.016 [-0.020, 0.050]	NaN	0.009 [-0.027, 0.044]
theta	NaN	NaN	-0.017 [-0.380, 0.562]	NaN
theta_0	NaN	NaN	NaN	0.244 [0.068, 0.477]
gamma	NaN	NaN	NaN	-0.006 [-0.045, 0.034]
rho_1	NaN	0.744 [0.370, 1.051]	0.899 [0.612, 1.112]	0.986 [0.767, 1.203]
rho_2	NaN	-0.026 [-0.309, 0.239]	0.015 [-0.179, 0.245]	-0.041 [-0.252, 0.168]
phi_1	0.632 [0.498, 0.768]	0.632 [0.499, 0.767]	0.632 [0.505, 0.753]	0.632 [0.494, 0.765]
lambda_ez	0.010 [-0.202, 0.218]	0.017 [-0.193, 0.224]	NaN	0.018 [-0.181, 0.215]
n	NaN	-0.022 [-0.084, 0.038]	-0.033 [-0.080, 0.021]	-0.001 [-0.053, 0.051]

12.2 Prior vs Posterior by Model

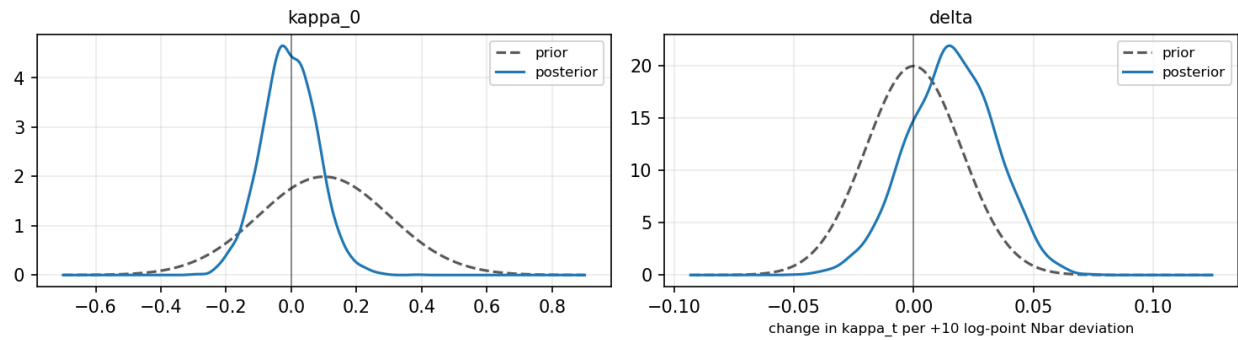
ces

Prior vs posterior — ces



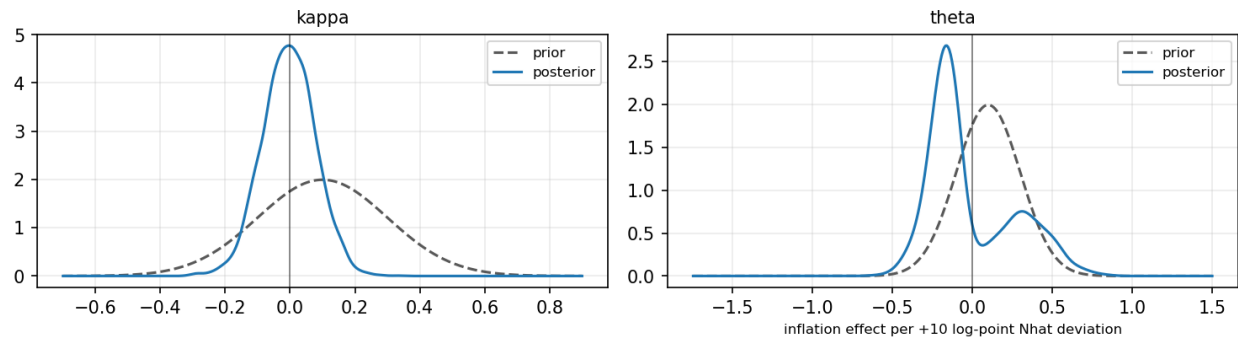
hsa_steady

Prior vs posterior — hsa_steady



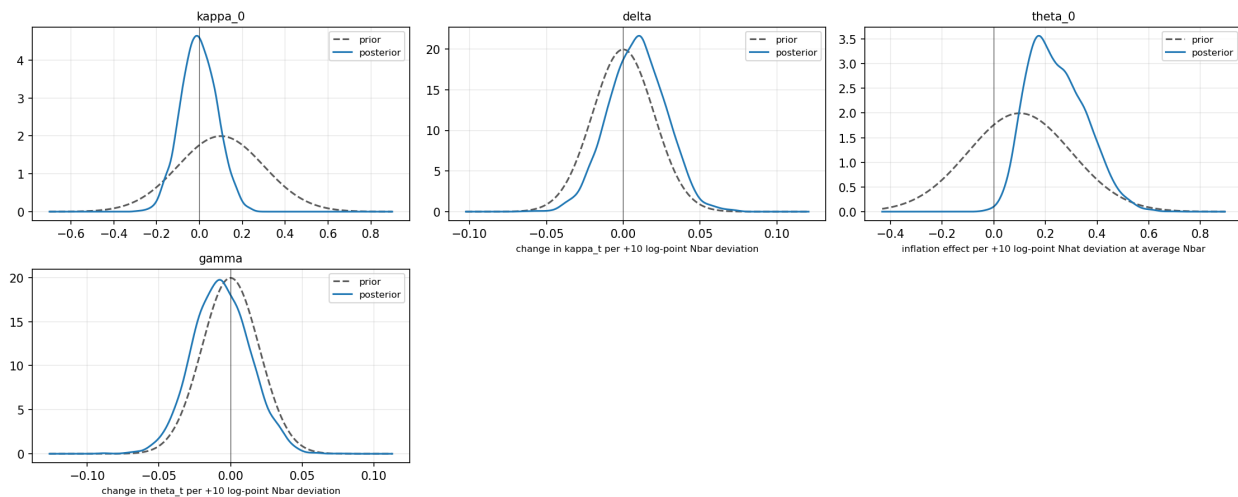
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

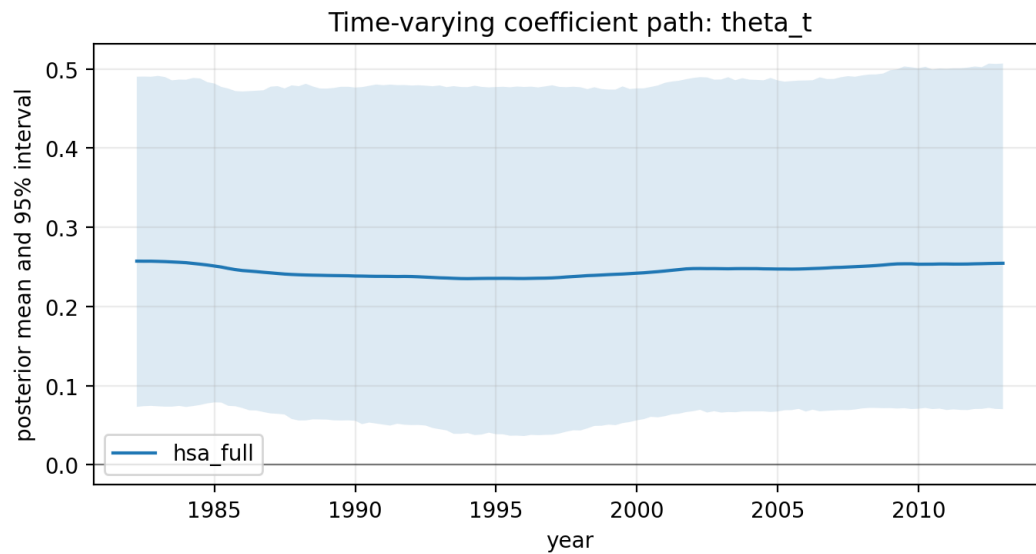
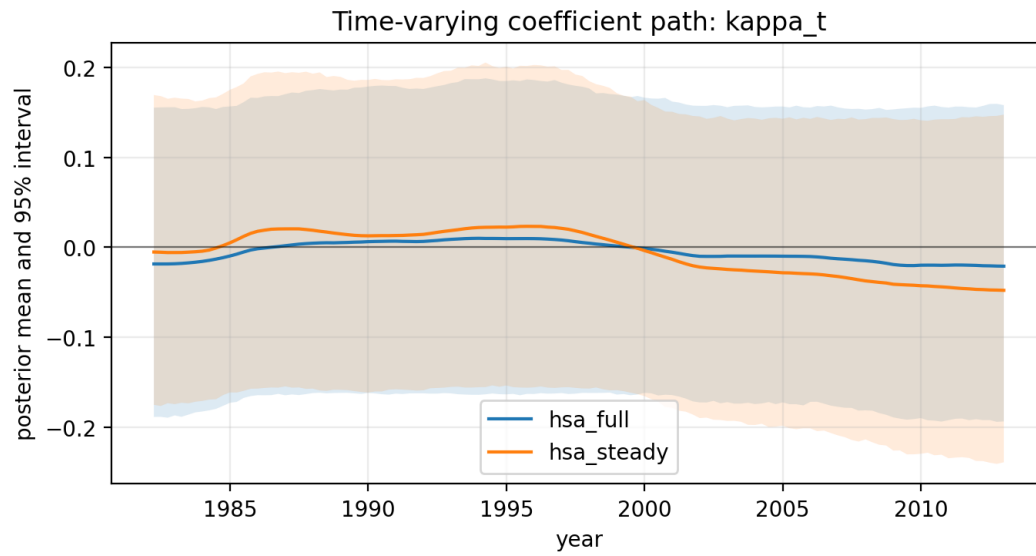
Prior vs posterior — hsa_full



12.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	2.7022
hsa_dynamic	kappa=0	0.1000	0.2000	2.7107
hsa_dynamic	theta=0	0.1000	0.2000	0.3408
hsa_full	kappa_0=0	0.1000	0.2000	2.6073
hsa_full	delta=0	0.0000	0.0200	0.9375
hsa_full	theta_0=0	0.1000	0.2000	0.0594
hsa_full	gamma=0	0.0000	0.0200	0.9031
hsa_steady	kappa_0=0	0.1000	0.2000	2.5211
hsa_steady	delta=0	0.0000	0.0200	0.7424

12.4 Time-Varying Kappa Path



12.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-121.9287	0.6455	-295.1854	1.0000	NaN	NaN	NaN
hsa_dynamic	-121.6114	0.6445	NaN	NaN	NaN	0.3408	NaN
hsa_full	-111.7415	0.5927	NaN	NaN	0.9375	NaN	0.9031
hsa_steady	-121.1657	0.6414	-356.8291	0.0000	0.7424	NaN	NaN

13 labor_share_gap_hp: prior=baseline, period=full, constraint=kappa i= 0

Condition. data=labor_share_gap_hp, prior=baseline, period=full, constraint=restricted_kappa (kappa i= 0).

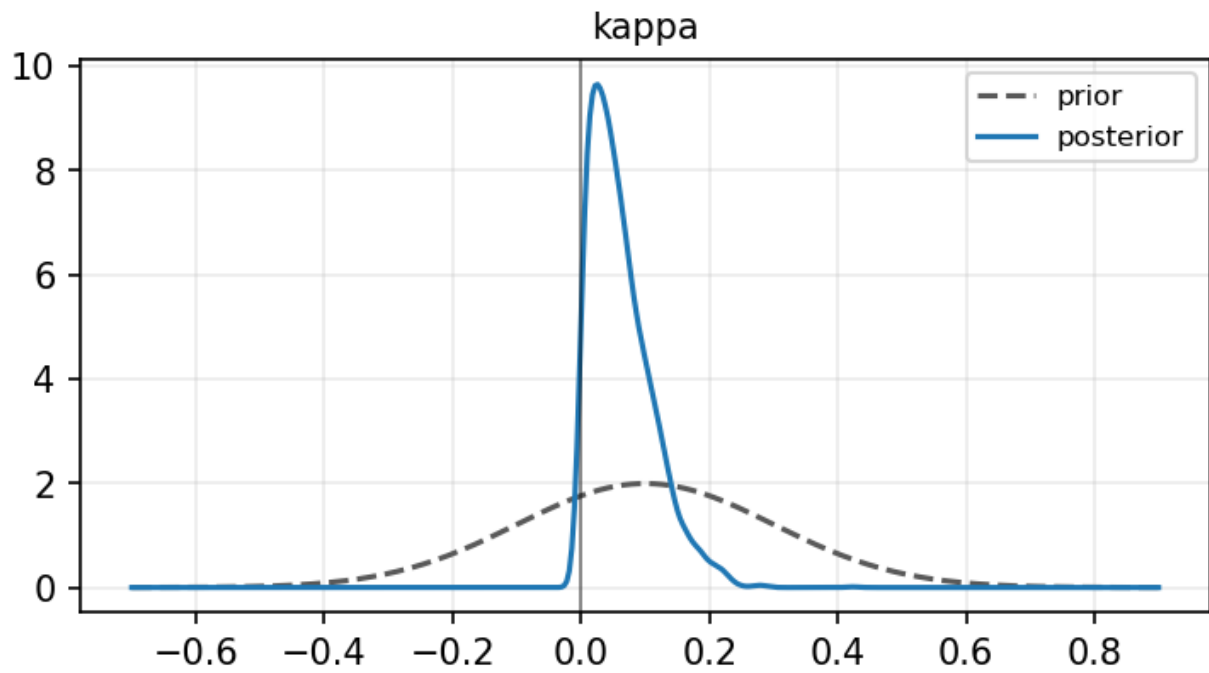
13.1 Coefficient Table

	ces	hsa_steady	hsa_dynamic	hsa_full
parameter				
alpha	0.629 [0.523, 0.736]	0.640 [0.537, 0.742]	0.610 [0.505, 0.715]	0.587 [0.480, 0.699]
kappa	0.063 [0.002, 0.183]	NaN	0.064 [0.002, 0.184]	NaN
kappa_0	NaN	-0.006 [-0.173, 0.157]	NaN	-0.003 [-0.165, 0.164]
delta	NaN	0.016 [-0.020, 0.050]	NaN	0.009 [-0.027, 0.044]
theta	NaN	NaN	-0.060 [-0.380, 0.541]	NaN
theta_0	NaN	NaN	NaN	0.244 [0.068, 0.477]
gamma	NaN	NaN	NaN	-0.006 [-0.045, 0.034]
rho_1	NaN	0.744 [0.370, 1.051]	0.905 [0.634, 1.121]	0.986 [0.767, 1.203]
rho_2	NaN	-0.026 [-0.309, 0.239]	0.011 [-0.195, 0.231]	-0.041 [-0.252, 0.168]
phi_1	0.632 [0.498, 0.763]	0.632 [0.499, 0.767]	0.630 [0.503, 0.763]	0.632 [0.494, 0.765]
lambda_ez	-0.059 [-0.229, 0.098]	0.017 [-0.193, 0.224]	NaN	0.018 [-0.181, 0.215]
n	NaN	-0.022 [-0.084, 0.038]	-0.036 [-0.082, 0.019]	-0.001 [-0.053, 0.051]

13.2 Prior vs Posterior by Model

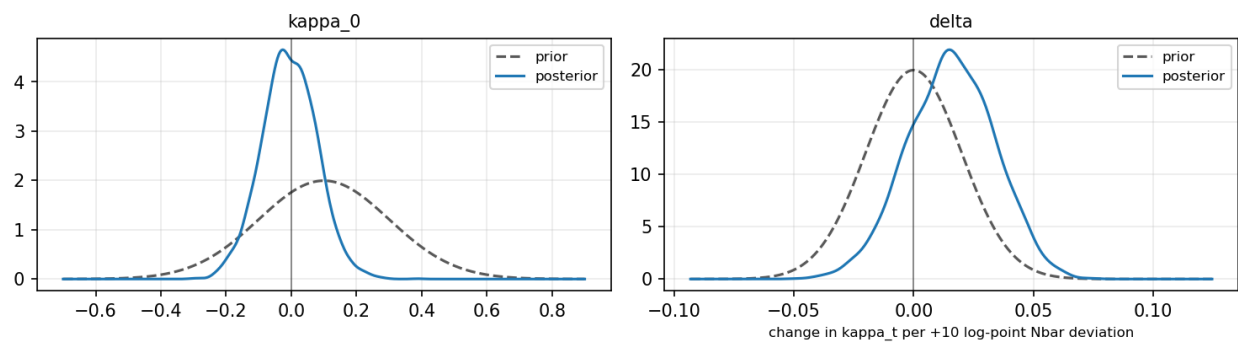
ces

Prior vs posterior — ces



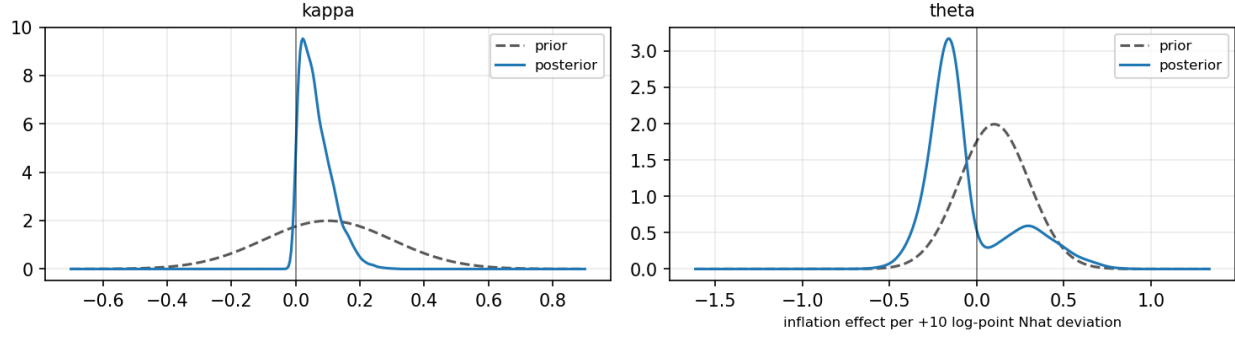
hsa_steady

Prior vs posterior — hsa_steady



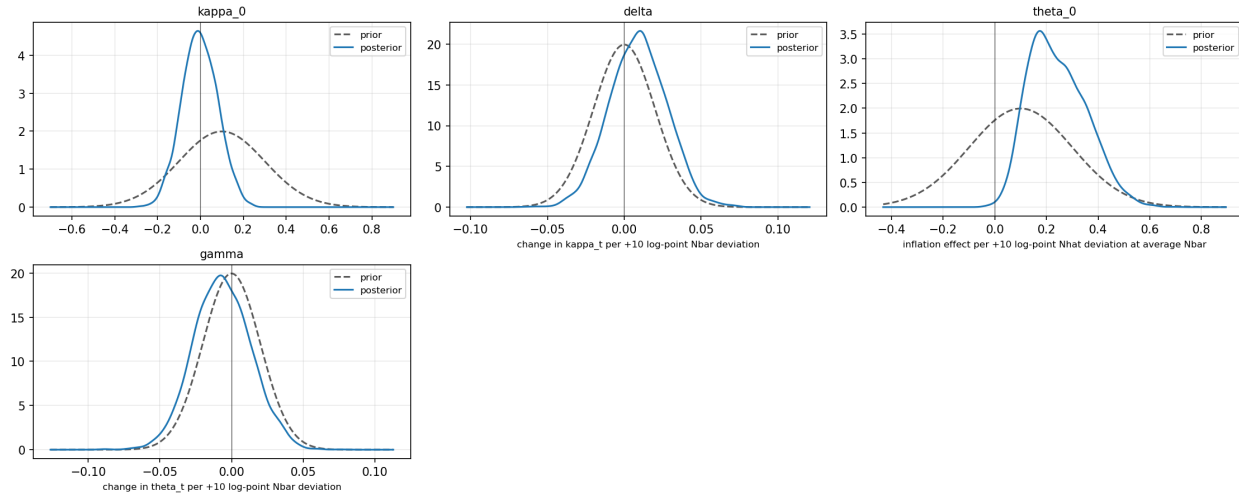
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

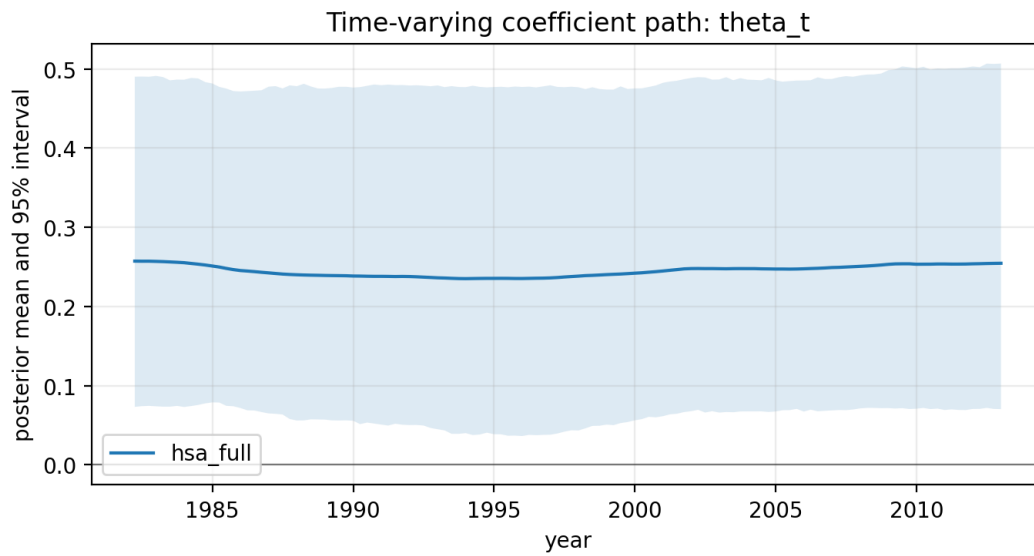
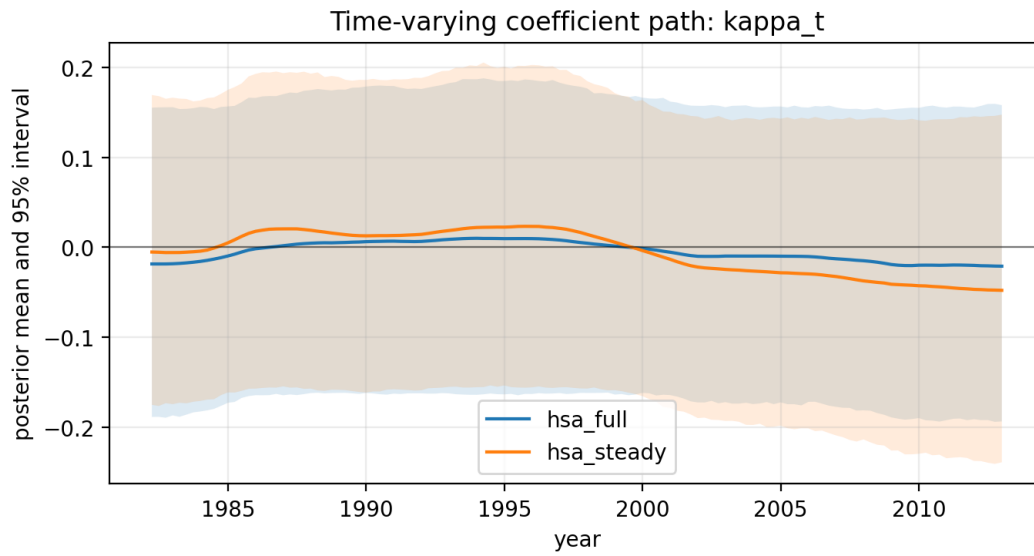
Prior vs posterior — hsa_full



13.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	2.8874
hsa_dynamic	kappa=0	0.1000	0.2000	2.7933
hsa_dynamic	theta=0	0.1000	0.2000	0.3011
hsa_full	kappa_0=0	0.1000	0.2000	2.6073
hsa_full	delta=0	0.0000	0.0200	0.9375
hsa_full	theta_0=0	0.1000	0.2000	0.0594
hsa_full	gamma=0	0.0000	0.0200	0.9031
hsa_steady	kappa_0=0	0.1000	0.2000	2.5211
hsa_steady	delta=0	0.0000	0.0200	0.7424

13.4 Time-Varying Kappa Path



13.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-122.7763	0.6503	-295.5721	1.0000	NaN	NaN	NaN
hsa_dynamic	-121.0586	0.6418	NaN	NaN	NaN	0.3011	NaN
hsa_full	-111.7415	0.5927	NaN	NaN	0.9375	NaN	0.9031
hsa_steady	-121.1657	0.6414	-356.8291	0.0000	0.7424	NaN	NaN

14 output_gap_bn: prior=baseline, period=full

Condition. data=output_gap_bn, prior=baseline, period=full, constraint=unrestricted (unrestricted).

14.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.6722	0.5564	0.7846
ces	kappa	0.0586	-0.0250	0.1440
hsa_steady	alpha	0.6573	0.5412	0.7708
hsa_steady	kappa_0	0.0461	-0.0428	0.1326
hsa_steady	delta	-0.0229	-0.0559	0.0105
hsa_dynamic	alpha	0.6362	0.5108	0.7583
hsa_dynamic	kappa	0.0838	-0.0094	0.1807
hsa_dynamic	theta	0.1174	-0.3556	0.6462
hsa_full	alpha	0.6043	0.4843	0.7254
hsa_full	kappa_0	0.0877	-0.0088	0.1791
hsa_full	delta	-0.0219	-0.0513	0.0148
hsa_full	theta_0	0.0755	-0.2828	0.5375
hsa_full	gamma	0.0008	-0.0361	0.0331

14.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

14.3 Savage-Dickey Density Ratio

note

No output available for this block.

14.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

14.5 Model Comparison

note

No output available for this block.

15 output_gap_bn: prior=baseline, period=pre_2008

Condition. data=output_gap_bn, prior=baseline, period=pre_2008, constraint=unrestricted (unrestricted).

15.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.7082	0.5706	0.8422
ces	kappa	0.0503	-0.0416	0.1419
hsa_steady	alpha	0.7057	0.5728	0.8384
hsa_steady	kappa_0	0.0449	-0.0538	0.1419
hsa_steady	delta	-0.0188	-0.0570	0.0206
hsa_dynamic	alpha	0.6245	0.4886	0.7657
hsa_dynamic	kappa	0.0686	-0.0215	0.1617
hsa_dynamic	theta	0.3838	-0.2853	0.7256
hsa_full	alpha	0.6261	0.4768	0.7778
hsa_full	kappa_0	0.0588	-0.0454	0.1632
hsa_full	delta	-0.0089	-0.0439	0.0254
hsa_full	theta_0	0.3229	0.0862	0.5826
hsa_full	gamma	-0.0055	-0.0432	0.0339

15.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

15.3 Savage-Dickey Density Ratio

note

No output available for this block.

15.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

15.5 Model Comparison

note

No output available for this block.

16 output_gap_bn: prior=baseline, period=post_2008

Condition. data=output_gap_bn, prior=baseline, period=post_2008, constraint=unrestricted (unrestricted).

16.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.5343	0.2877	0.7757
ces	kappa	0.1707	-0.0498	0.3839
hsa_steady	alpha	0.5378	0.2990	0.7758
hsa_steady	kappa_0	0.1669	-0.0538	0.3776
hsa_steady	delta	0.0008	-0.0398	0.0399
hsa_dynamic	alpha	0.5501	0.3266	0.7676
hsa_dynamic	kappa	0.1553	-0.0464	0.3561
hsa_dynamic	theta	0.1200	-0.3065	0.5508
hsa_full	alpha	0.5356	0.2868	0.7671
hsa_full	kappa_0	0.1695	-0.0413	0.3902
hsa_full	delta	0.0009	-0.0380	0.0414
hsa_full	theta_0	0.0912	-0.2895	0.4820
hsa_full	gamma	0.0004	-0.0386	0.0388

16.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

16.3 Savage-Dickey Density Ratio

note

No output available for this block.

16.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

16.5 Model Comparison

note

No output available for this block.

17 output_gap_bn: prior=baseline, period=exclude_covid

Condition. data=output_gap_bn, prior=baseline, period=exclude_covid, constraint=unrestricted (unrestricted).

17.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.6709	0.5519	0.7871
ces	kappa	0.0600	-0.0247	0.1435
hsa_steady	alpha	0.6588	0.5418	0.7785
hsa_steady	kappa_0	0.0472	-0.0454	0.1380
hsa_steady	delta	-0.0230	-0.0561	0.0110
hsa_dynamic	alpha	0.6294	0.5076	0.7499
hsa_dynamic	kappa	0.0906	-0.0021	0.1849
hsa_dynamic	theta	0.2118	-0.3027	0.6683
hsa_full	alpha	0.6144	0.4963	0.7297
hsa_full	kappa_0	0.0960	0.0000	0.1911
hsa_full	delta	-0.0073	-0.0463	0.0315
hsa_full	theta_0	0.2333	0.0659	0.5118
hsa_full	gamma	-0.0100	-0.0459	0.0296

17.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

17.3 Savage-Dickey Density Ratio

note

No output available for this block.

17.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

17.5 Model Comparison

note

No output available for this block.

18 output_gap_bn: prior=baseline, period=start_1988

Condition. data=output_gap_bn, prior=baseline, period=start_1988, constraint=unrestricted (unrestricted).

18.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.6412	0.5150	0.7745
ces	kappa	0.0834	-0.0161	0.1860
hsa_steady	alpha	0.6306	0.5031	0.7557
hsa_steady	kappa_0	0.0679	-0.0382	0.1721
hsa_steady	delta	-0.0184	-0.0510	0.0157
hsa_dynamic	alpha	0.6016	0.4661	0.7340
hsa_dynamic	kappa	0.1310	0.0102	0.2587
hsa_dynamic	theta	0.2762	-0.1591	0.6295
hsa_full	alpha	0.5699	0.4324	0.7097
hsa_full	kappa_0	0.1404	0.0182	0.2742
hsa_full	delta	-0.0190	-0.0539	0.0159
hsa_full	theta_0	0.2839	0.0512	0.5312
hsa_full	gamma	-0.0053	-0.0443	0.0330

18.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

18.3 Savage-Dickey Density Ratio

note

No output available for this block.

18.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

18.5 Model Comparison

note

No output available for this block.

19 output_gap_bn: prior=baseline, period=end_2019

Condition. data=output_gap_bn, prior=baseline, period=end_2019, constraint=unrestricted (unrestricted).

19.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.6724	0.5545	0.7852
ces	kappa	0.0591	-0.0237	0.1418
hsa_steady	alpha	0.6604	0.5449	0.7740
hsa_steady	kappa_0	0.0476	-0.0445	0.1388
hsa_steady	delta	-0.0228	-0.0561	0.0102
hsa_dynamic	alpha	0.6290	0.5079	0.7472
hsa_dynamic	kappa	0.0906	-0.0002	0.1856
hsa_dynamic	theta	0.1933	-0.3264	0.6561
hsa_full	alpha	0.6179	0.4905	0.7380
hsa_full	kappa_0	0.0831	-0.0128	0.1849
hsa_full	delta	-0.0201	-0.0522	0.0143
hsa_full	theta_0	0.0932	-0.3328	0.5851
hsa_full	gamma	-0.0025	-0.0432	0.0349

19.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

19.3 Savage-Dickey Density Ratio

note

No output available for this block.

19.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

19.5 Model Comparison

note

No output available for this block.

20 output_gap_hp: prior=baseline, period=full

Condition. data=output_gap_hp, prior=baseline, period=full, constraint=unrestricted (unrestricted).

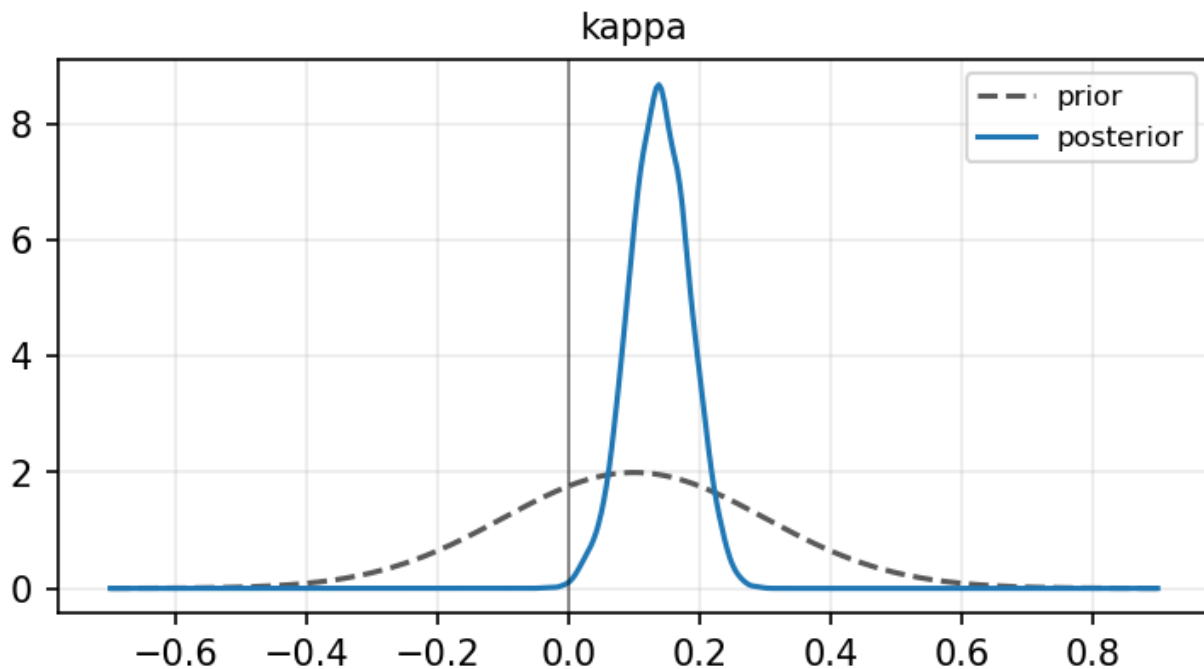
20.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.670 [0.566, 0.773]	0.664 [0.561, 0.762]	0.637 [0.519, 0.750]	0.624 [0.507, 0.739]
kappa	0.138 [0.047, 0.227]	NaN	0.144 [0.047, 0.239]	NaN
kappa_0	NaN	0.135 [0.042, 0.225]	NaN	0.139 [0.035, 0.239]
delta	NaN	-0.011 [-0.044, 0.024]	NaN	-0.011 [-0.047, 0.024]
theta	NaN	NaN	0.168 [-0.216, 0.592]	NaN
theta_0	NaN	NaN	NaN	0.220 [0.015, 0.518]
gamma	NaN	NaN	NaN	-0.003 [-0.043, 0.036]
rho_1	NaN	0.755 [0.403, 1.044]	0.878 [0.507, 1.116]	0.967 [0.701, 1.208]
rho_2	NaN	-0.018 [-0.300, 0.233]	0.005 [-0.202, 0.216]	-0.051 [-0.276, 0.177]
phi_1	0.896 [0.821, 0.972]	0.895 [0.822, 0.968]	0.897 [0.827, 0.967]	0.897 [0.825, 0.972]
lambda_ez	0.134 [-0.093, 0.361]	0.127 [-0.098, 0.351]	NaN	0.099 [-0.131, 0.326]
n	NaN	-0.020 [-0.083, 0.041]	-0.021 [-0.070, 0.025]	-0.008 [-0.061, 0.044]

20.2 Prior vs Posterior by Model

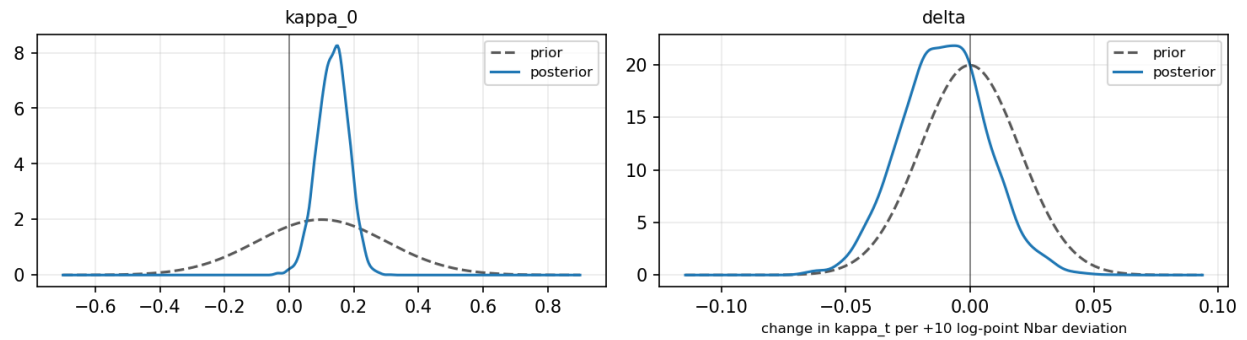
ces

Prior vs posterior — ces



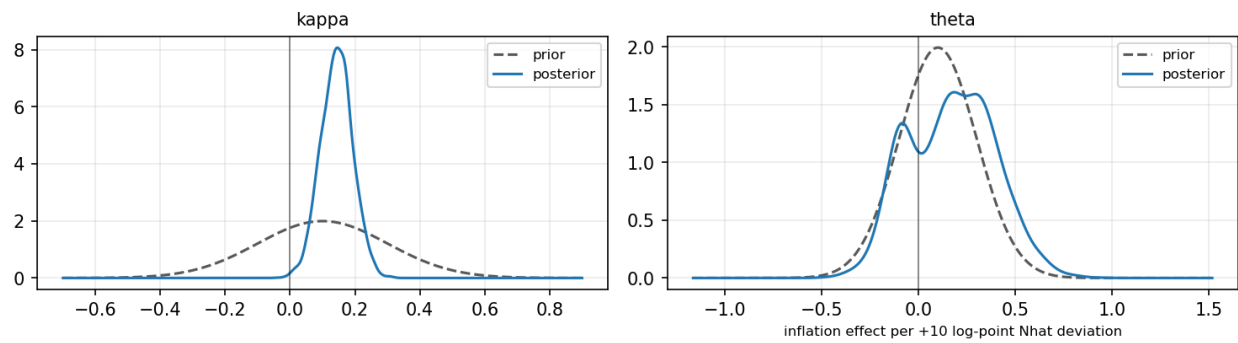
hsa_steady

Prior vs posterior — hsa_steady



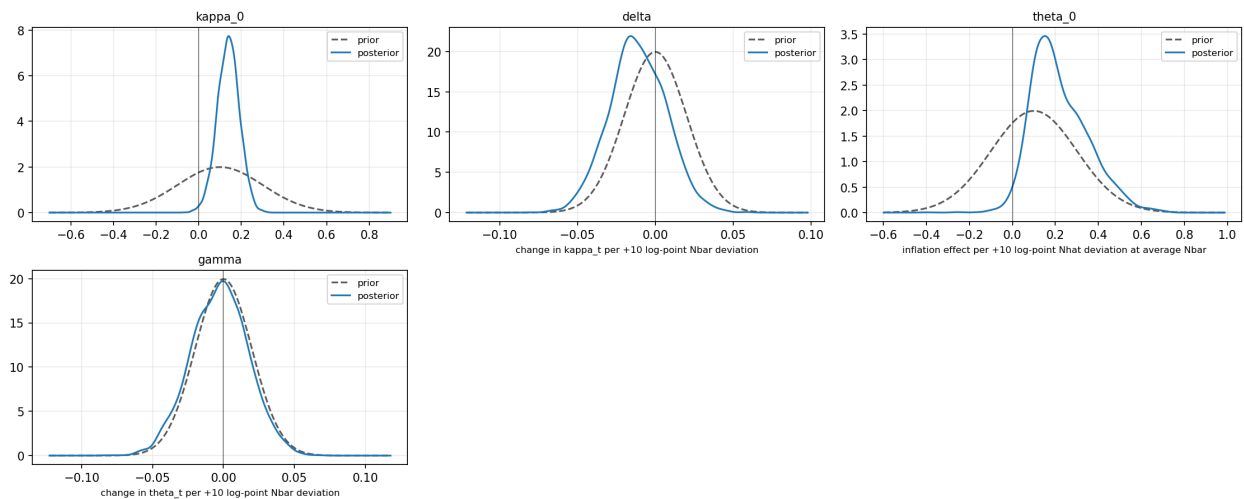
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

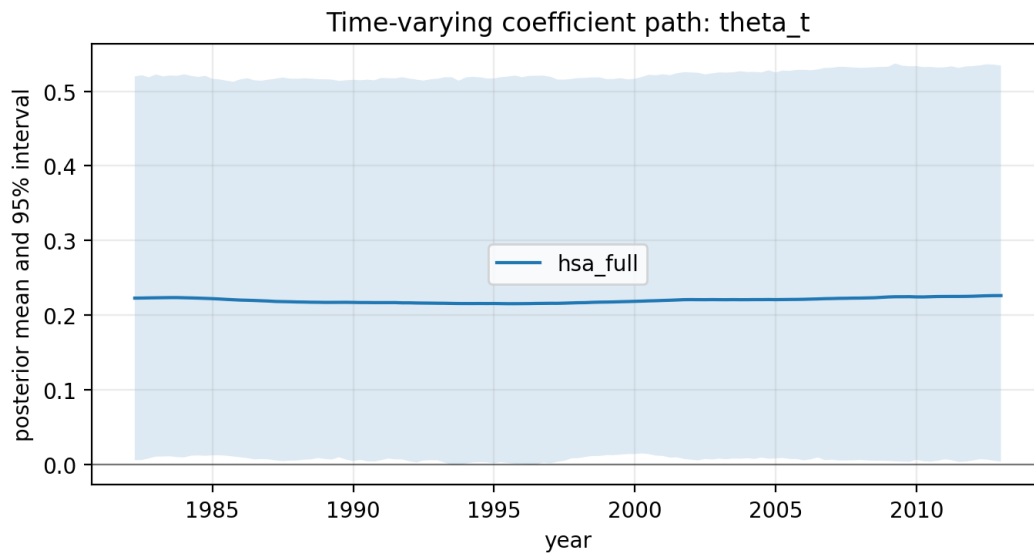
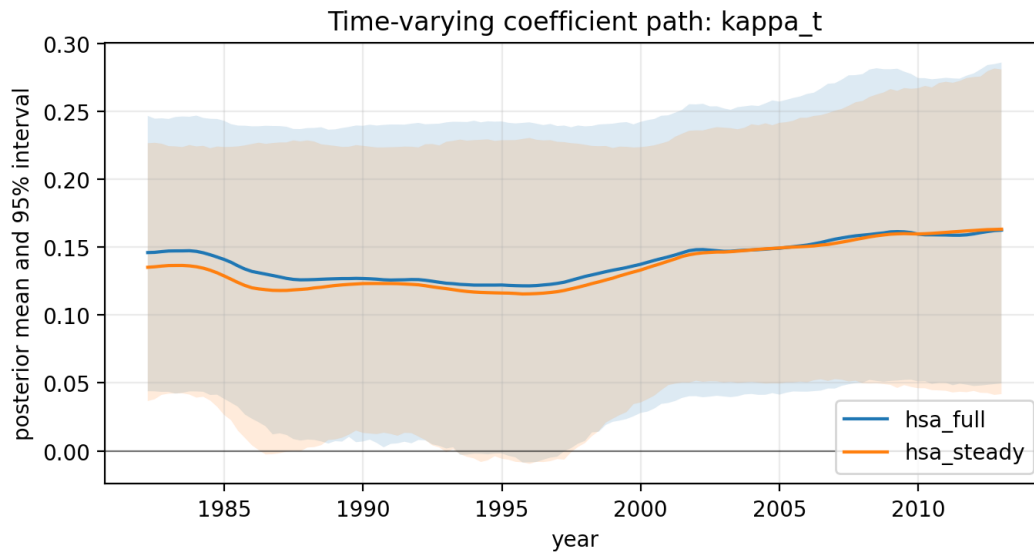
Prior vs posterior — hsa_full



20.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	0.0587
hsa_dynamic	kappa=0	0.1000	0.2000	0.1019
hsa_dynamic	theta=0	0.1000	0.2000	0.6200
hsa_full	kappa_0=0	0.1000	0.2000	0.1666
hsa_full	delta=0	0.0000	0.0200	0.8608
hsa_full	theta_0=0	0.1000	0.2000	0.2893
hsa_full	gamma=0	0.0000	0.0200	0.9883
hsa_steady	kappa_0=0	0.1000	0.2000	0.1246
hsa_steady	delta=0	0.0000	0.0200	1.0043

20.4 Time-Varying Kappa Path



20.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-114.0771	0.6059	-239.3121	1.0000	NaN	NaN	NaN
hsa_dynamic	-111.6291	0.5953	NaN	NaN	NaN	0.6200	NaN
hsa_full	-105.9814	0.5657	NaN	NaN	0.8608	NaN	0.9883
hsa_steady	-113.6351	0.6035	-302.1031	0.0000	1.0043	NaN	NaN

21 output_gap_hp: prior=baseline, period=full, constraint=kappa $i = 0$

Condition. data=output_gap_hp, prior=baseline, period=full, constraint=restricted_kappa (kappa $i = 0$).

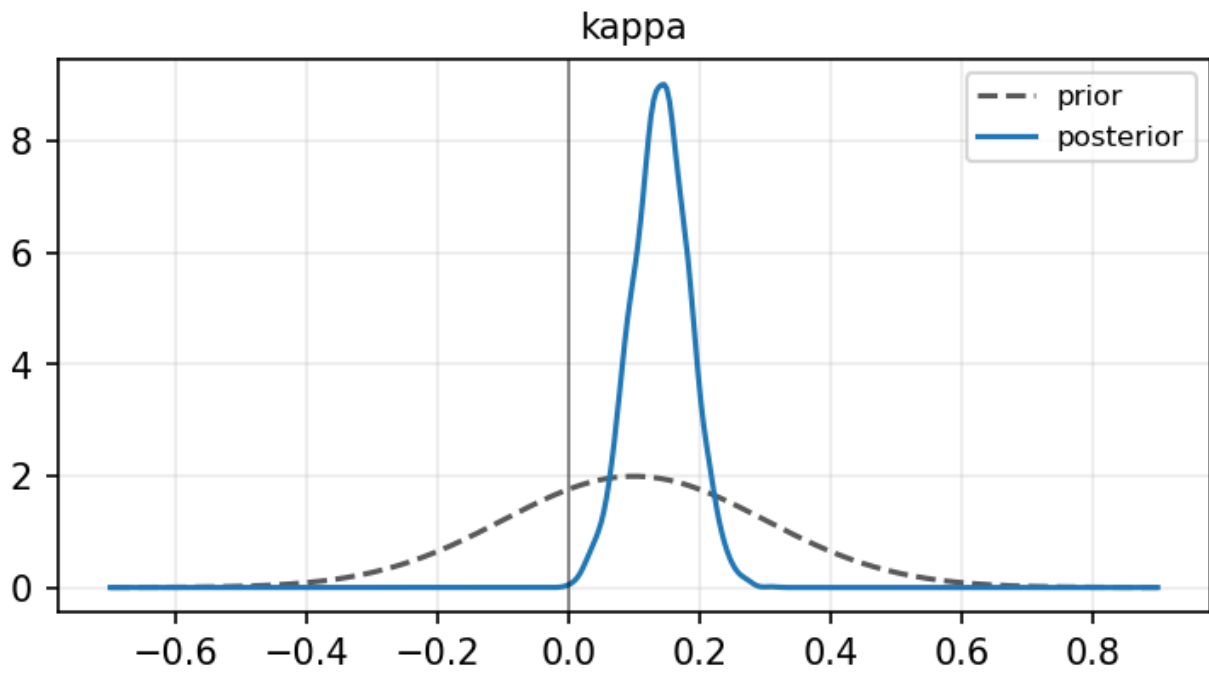
21.1 Coefficient Table

	ces	hsa_steady	hsa_dynamic	hsa_full
parameter				
alpha	0.672 [0.564, 0.776]	0.664 [0.561, 0.762]	0.639 [0.523, 0.747]	0.624 [0.507, 0.739]
kappa	0.140 [0.052, 0.227]	NaN	0.141 [0.050, 0.238]	NaN
kappa_0	NaN	0.135 [0.042, 0.225]	NaN	0.139 [0.035, 0.239]
delta	NaN	-0.011 [-0.044, 0.024]	NaN	-0.011 [-0.047, 0.024]
theta	NaN	NaN	0.149 [-0.223, 0.579]	NaN
theta_0	NaN	NaN	NaN	0.220 [0.015, 0.518]
gamma	NaN	NaN	NaN	-0.003 [-0.043, 0.036]
rho_1	NaN	0.755 [0.403, 1.044]	0.887 [0.524, 1.123]	0.967 [0.701, 1.208]
rho_2	NaN	-0.018 [-0.300, 0.233]	0.005 [-0.218, 0.216]	-0.051 [-0.276, 0.177]
phi_1	0.896 [0.821, 0.972]	0.895 [0.822, 0.968]	0.898 [0.831, 0.969]	0.897 [0.825, 0.972]
lambda_ez	0.131 [-0.093, 0.358]	0.127 [-0.098, 0.351]	NaN	0.099 [-0.131, 0.326]
n	NaN	-0.020 [-0.083, 0.041]	-0.023 [-0.072, 0.025]	-0.008 [-0.061, 0.044]

21.2 Prior vs Posterior by Model

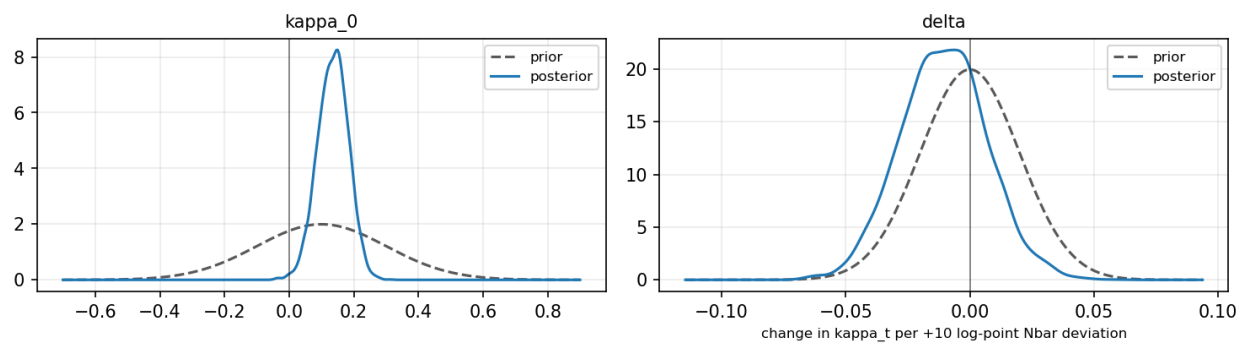
ces

Prior vs posterior — ces



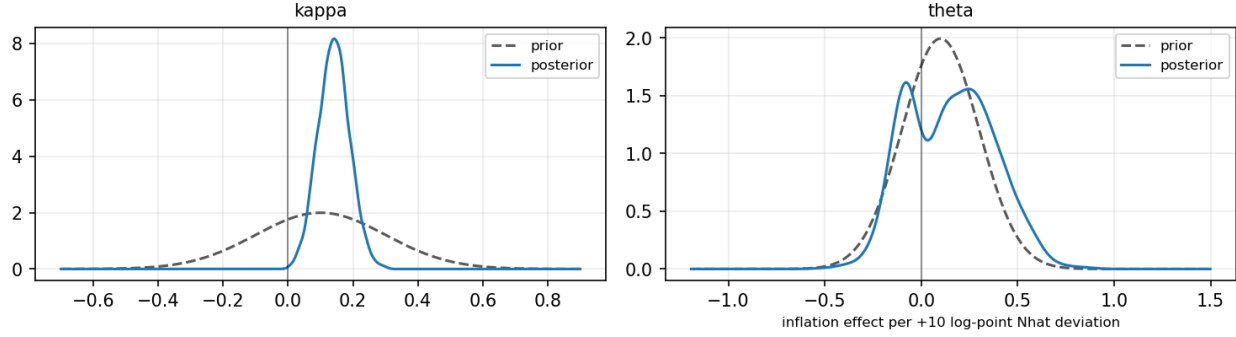
hsa_steady

Prior vs posterior — hsa_steady



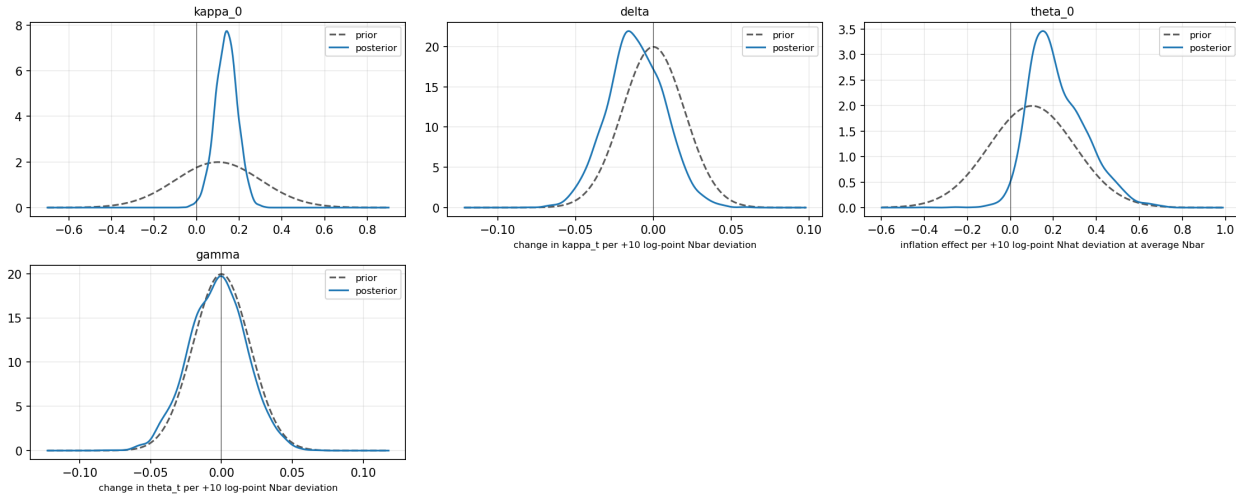
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

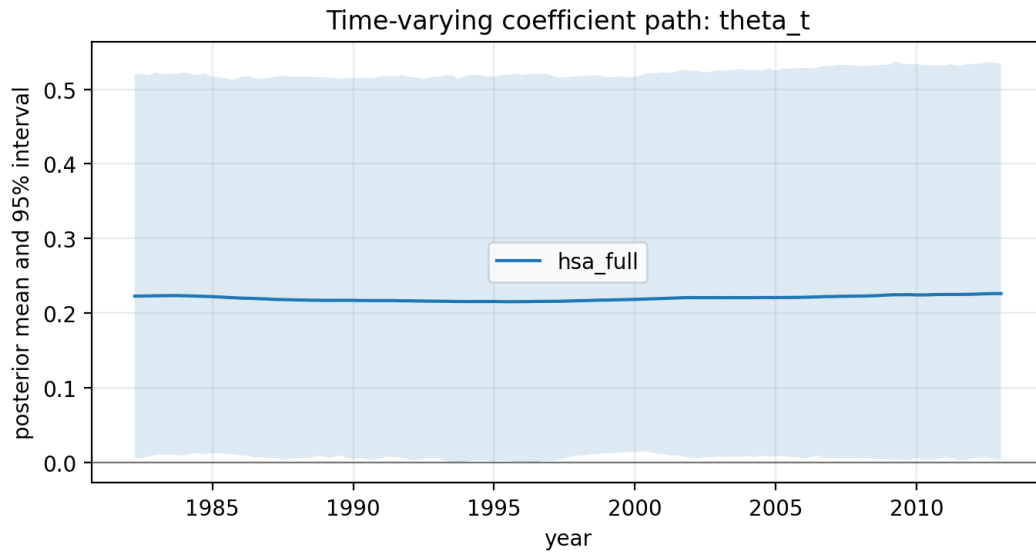
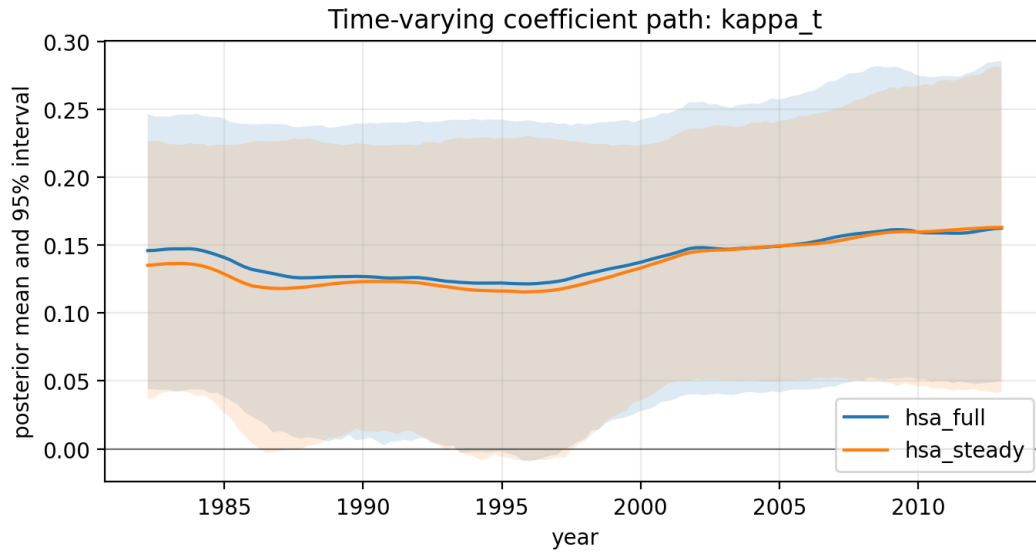
Prior vs posterior — hsa_full



21.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	0.0282
hsa_dynamic	kappa=0	0.1000	0.2000	0.0459
hsa_dynamic	theta=0	0.1000	0.2000	0.6847
hsa_full	kappa_0=0	0.1000	0.2000	0.1666
hsa_full	delta=0	0.0000	0.0200	0.8608
hsa_full	theta_0=0	0.1000	0.2000	0.2893
hsa_full	gamma=0	0.0000	0.0200	0.9883
hsa_steady	kappa_0=0	0.1000	0.2000	0.1246
hsa_steady	delta=0	0.0000	0.0200	1.0043

21.4 Time-Varying Kappa Path



21.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-114.0810	0.6058	-239.3093	1.0000	NaN	NaN	NaN
hsa_dynamic	-112.3597	0.5987	NaN	NaN	NaN	0.6847	NaN
hsa_full	-105.9814	0.5657	NaN	NaN	0.8608	NaN	0.9883
hsa_steady	-113.6351	0.6035	-302.1031	0.0000	1.0043	NaN	NaN

22 unemployment_gap: prior=baseline, period=full

Condition. data=unemployment_gap, prior=baseline, period=full, constraint=unrestricted (unrestricted).

22.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.6470	0.5434	0.7494
ces	kappa	0.0556	-0.0075	0.1200
hsa_steady	alpha	0.6411	0.5429	0.7417
hsa_steady	kappa_0	0.0913	0.0126	0.1757
hsa_steady	delta	0.0287	0.0006	0.0556
hsa_dynamic	alpha	0.5936	0.4743	0.7085
hsa_dynamic	kappa	0.0849	-0.0113	0.1913
hsa_dynamic	theta	0.2369	-0.2658	0.6191
hsa_full	alpha	0.6126	0.4845	0.7309
hsa_full	kappa_0	0.0766	-0.0114	0.1738
hsa_full	delta	0.0183	-0.0263	0.0520
hsa_full	theta_0	0.1198	-0.2058	0.4939
hsa_full	gamma	-0.0059	-0.0443	0.0346

22.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

22.3 Savage-Dickey Density Ratio

note

No output available for this block.

22.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

22.5 Model Comparison

note

No output available for this block.

23 unemployment_gap: prior=baseline, period=pre_2008

Condition. data=unemployment_gap, prior=baseline, period=pre_2008, constraint=unrestricted (unrestricted).

23.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.6804	0.5720	0.7828
ces	kappa	0.1649	0.0865	0.2433
hsa_steady	alpha	0.6817	0.5760	0.7871
hsa_steady	kappa_0	0.1619	0.0751	0.2481
hsa_steady	delta	0.0039	-0.0345	0.0428
hsa_dynamic	alpha	0.6327	0.5034	0.7530
hsa_dynamic	kappa	0.1803	0.0745	0.2904
hsa_dynamic	theta	0.2923	-0.1692	0.6800
hsa_full	alpha	0.6140	0.4869	0.7424
hsa_full	kappa_0	0.1778	0.0523	0.3115
hsa_full	delta	0.0015	-0.0345	0.0367
hsa_full	theta_0	0.2595	0.0187	0.5258
hsa_full	gamma	-0.0000	-0.0397	0.0379

23.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

23.3 Savage-Dickey Density Ratio

note

No output available for this block.

23.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

23.5 Model Comparison

note

No output available for this block.

24 unemployment_gap: prior=baseline, period=post_2008

Condition. data=unemployment_gap, prior=baseline, period=post_2008, constraint=unrestricted (unrestricted).

24.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.5728	0.3258	0.8074
ces	kappa	-0.0185	-0.1491	0.1077
hsa_steady	alpha	0.5752	0.3284	0.8102
hsa_steady	kappa_0	-0.0187	-0.1414	0.1105
hsa_steady	delta	0.0011	-0.0397	0.0423
hsa_dynamic	alpha	0.5878	0.3651	0.8122
hsa_dynamic	kappa	-0.0222	-0.1334	0.0860
hsa_dynamic	theta	0.1581	-0.3597	0.6228
hsa_full	alpha	0.5744	0.3238	0.8026
hsa_full	kappa_0	-0.0218	-0.1473	0.1087
hsa_full	delta	0.0011	-0.0387	0.0412
hsa_full	theta_0	0.1039	-0.3071	0.5091
hsa_full	gamma	-0.0005	-0.0398	0.0383

24.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

24.3 Savage-Dickey Density Ratio

note

No output available for this block.

24.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

24.5 Model Comparison

note

No output available for this block.

25 unemployment_gap: prior=baseline, period=exclude_covid

Condition. data=unemployment_gap, prior=baseline, period=exclude_covid, constraint=unrestricted (unrestricted).

25.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.6452	0.5385	0.7491
ces	kappa	0.0567	-0.0080	0.1196
hsa_steady	alpha	0.6420	0.5394	0.7376
hsa_steady	kappa_0	0.0913	0.0107	0.1786
hsa_steady	delta	0.0287	-0.0000	0.0564
hsa_dynamic	alpha	0.5937	0.4759	0.7108
hsa_dynamic	kappa	0.0809	-0.0136	0.1897
hsa_dynamic	theta	0.2200	-0.2847	0.6272
hsa_full	alpha	0.5939	0.4652	0.7152
hsa_full	kappa_0	0.0919	-0.0066	0.1891
hsa_full	delta	0.0115	-0.0342	0.0519
hsa_full	theta_0	0.2128	-0.1005	0.5312
hsa_full	gamma	-0.0036	-0.0425	0.0349

25.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

25.3 Savage-Dickey Density Ratio

note

No output available for this block.

25.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

25.5 Model Comparison

note

No output available for this block.

26 unemployment_gap: prior=baseline, period=start_1988

Condition. data=unemployment_gap, prior=baseline, period=start_1988, constraint=unrestricted (unrestricted).

26.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.6262	0.5024	0.7461
ces	kappa	-0.0067	-0.0808	0.0699
hsa_steady	alpha	0.6266	0.5062	0.7457
hsa_steady	kappa_0	0.0079	-0.0827	0.1017
hsa_steady	delta	0.0075	-0.0241	0.0405
hsa_dynamic	alpha	0.6157	0.4921	0.7356
hsa_dynamic	kappa	0.0002	-0.0742	0.0839
hsa_dynamic	theta	0.1560	-0.2522	0.5550
hsa_full	alpha	0.6081	0.4762	0.7312
hsa_full	kappa_0	0.0145	-0.0814	0.1126
hsa_full	delta	0.0053	-0.0300	0.0372
hsa_full	theta_0	0.1520	-0.2110	0.5322
hsa_full	gamma	-0.0007	-0.0395	0.0384

26.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

26.3 Savage-Dickey Density Ratio

note

No output available for this block.

26.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

26.5 Model Comparison

note

No output available for this block.

27 unemployment_gap: prior=baseline, period=end_2019

Condition. data=unemployment_gap, prior=baseline, period=end_2019, constraint=unrestricted (unrestricted).

27.1 Coefficient Table

model	parameter	posterior_mean	ci_2.5	ci_97.5
ces	alpha	0.6469	0.5417	0.7490
ces	kappa	0.0560	-0.0074	0.1193
hsa_steady	alpha	0.6401	0.5393	0.7400
hsa_steady	kappa_0	0.0897	0.0108	0.1759
hsa_steady	delta	0.0291	0.0013	0.0562
hsa_dynamic	alpha	0.6006	0.4836	0.7129
hsa_dynamic	kappa	0.0759	-0.0224	0.1870
hsa_dynamic	theta	0.1743	-0.3031	0.6026
hsa_full	alpha	0.5858	0.4683	0.6983
hsa_full	kappa_0	0.0971	0.0078	0.1905
hsa_full	delta	0.0078	-0.0324	0.0476
hsa_full	theta_0	0.2787	-0.0676	0.5703
hsa_full	gamma	-0.0025	-0.0420	0.0380

27.2 Prior vs Posterior by Model

No prior/posterior figures available for this block.

27.3 Savage-Dickey Density Ratio

note

No output available for this block.

27.4 Time-Varying Kappa Path

No kappa_t path for this block.

No theta_t path for this block.

27.5 Model Comparison

note

No output available for this block.

28 Notes

WAIC and LOO are not used as primary criteria because the models contain latent states. Chib marginal likelihoods are reported only when the legacy ordinate calculation can be matched to the

sampler units and supplied data.